Kefan Xu

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Summary

I am a fifth-year CS HCI Ph.D. Candidate at the Georgia Institute of Technology, operating at the intersection of HCI, CSCW, and Health Informatics. Rooted in a commitment to health equity, my research develops human-centered systems as probes to facilitate sensemaking, coping, and dynamic management strategies for chronic conditions. I design and develop functional mobile applications evaluated through longitudinal deployment studies. Looking forward, I aim to explore how Human-Centered AI can further enhance situated care ecologies. As an educator, I embrace a multidisciplinary, student-centered teaching philosophy that bridges theoretical HCI concepts with practical software development, empowering mentees to tackle real-world health challenges and succeed in both academia and industry.

Education

2022 – Present **Georgia Institute of Technology** – Atlanta, GA
Ph.D. in Computer Science (HCI Track)
Advisor: Dr. Rosa I. Arriaga. GPA: 4.0/4.0.

2019 – 2021 **University of Michigan** – Ann Arbor, MI
M.S. in Information Science
Advisor: Dr. Mark W. Newman. GPA: 4.0/4.0.

2015 – 2019 **New York University** – New York City, NY & Shanghai, China
B.S. in Data Science | B.S. in Interactive Media Arts (Doubled)
GPA: 3.6/4.0.

Honors and scholarships

2025 **OMSCS Pre-Doc Fellowship** (Georgia Institute of Technology: 3 Semesters)
2022 **Google Health Equity Research Initiative** (Google)
2021 **UMSI Master Thesis (MTOP) Grant** (University of Michigan)
2021 **IHPI/HSR Summer Student Fellowship** (University of Michigan)
2019 **NYU Founders Day Award** (New York University)
2016 **Tel Aviv University Summer Scholar Program** (Tel Aviv University)

Peer-reviewed Full Conference Papers

C7 2026 **Postphenomenological Insights from an In-the-Wild Collective Sensing System Deployment for PTSD Therapy**
Nathaniel Swinger, Myeonghan Ryu, Cynthia M. Baseman, **Kefan Xu**, Andrew M. Sherrill, Rosa I. Arriaga.
Proc. ACM Hum.-Comput. Interact. (CSCW '26).

C6 2026 **Human-centered Perspectives on a Clinical Decision Support System for Intensive Outpatient Veteran PTSD Care**
Cynthia M. Baseman, Myeonghan Ryu, Nathaniel Swinger, **Kefan Xu**, Andrew M. Sherrill, Rosa I. Arriaga.
Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26).

C5 2025 **Explainable AI for Daily Scenarios from End-Users' Perspective: Non-Use, Concerns, and Ideal Design**
Lingqing Wang, Chidimma Lois Anyi, **Kefan Xu**, Yifan Liu, Rosa I. Arriaga, Ashok K. Goel.
Proceedings of the 2025 ACM Designing Interactive Systems Conference (DIS '25).

C4 2025 **Understanding the Temporality of Informal Caregivers' Sense-Making on Conflicts and Life-Changing Events through Online Health Communities**
Kefan Xu, Cynthia M. Baseman, Nathaniel Swinger, Myeonghan Ryu, Rosa I. Arriaga.
Proc. ACM Hum.-Comput. Interact. (CSCW '25).

C3 2024 **Understanding the Effect of Reflective Iteration on Individuals' Physical Activity Planning**
Kefan Xu, Xinghui Yan, Myeonghan Ryu, Mark W. Newman, Rosa I. Arriaga.
Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24).

C2 2024 **Using Sensor-Captured Patient-Generated Data to Support Clinical Decision-making in PTSD Therapy**
Hayley I. Evans, Myeonghan Ryu, Theresa Hsieh, Jiawei Zhou, **Kefan Xu**, Kenneth W. Akers, Andrew M. Sherrill, Rosa I. Arriaga.
Proc. ACM Hum.-Comput. Interact. (CSCW '24).

C1 2022 **Understanding People's Experience for Physical Activity Planning and Exploring the Impact of Historical Records on Plan Creation and Execution**
Kefan Xu, Xinghui Yan, Mark W. Newman.
CHI Conference on Human Factors in Computing Systems (CHI '22).

Peer-reviewed Journal Articles

J1 2026 **"I'm a person—not a patient or diagnosis": Centering the Person Living with Dementia's Perspective on Caregiver Dynamics and Technology Design**
Hao Yuan, **Kefan Xu**, Rosa I. Arriaga, Jina Huh-Yoo.
ACM Transactions on Computing for Healthcare.

Workshop Papers

W2 2026 **Exploring How AI Supports Navigation of the Care Ecology During Life-Changing Events in Chronic Care**
Kefan Xu.
Doctoral Colloquium at the ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp / ISWC '26).

W1 2025 **Facilitating Individuals' Sensemaking about Sedentary Behavior via Contextualized Data**
Kefan Xu, Rosa I. Arriaga.
Workshop at the ACM CHI Conference on Human Factors in Computing Systems (CHI '25). arXiv:2509.19420.

Accepted / Under review

Evaluating Mobile-Captured Gait Visualizations with Experts and Rural Community Members Living with Diabetes
Ghahremani, S., Baseman, C.M., **Xu, K.**, Hawks, M.R., Arriaga, R.I.
Accepted with major revisions, ACM CHI Conference on Human Factors in Computing Systems (CHI '27).

Developmental Informatics in Care Ecologies: Navigating the Evolving Landscape for Chronic Care Management
Xu, K., Ghahremani, S., Thukral, A., Soneji, R., Fayfman, M., Schechter, C.M., Arriaga, R.I.
Accepted, Proc. ACM Hum.-Comput. Interact. (CSCW '27).

Understanding Between-Session Care and Designing AI-Assisted Clinician Tools to Facilitate Management Coherency
Xu, K., Wu, S., Liu, R., Wang, L., Arriaga, R.I., Adams, A.T.
Accepted with major revisions, ACM CHI Conference on Human Factors in Computing Systems (CHI '27).

Beyond a Number: How People Construct, Calibrate, and Contextualize Numeric Mental Self-Reports Over Time
Xu, K., Lee, J., Jung, J.,Thukral, A., Arriaga, R.I.
Accepted with major revisions, ACM CHI Conference on Human Factors in Computing Systems (CHI '27).

Beyond the Prompt Box: Co-designing Multimodal Generative AI with University Students in Everyday Contexts
Zhang, C., Huang, J., **Xu, K.**, Magerko, B.
In submission.

Projects

2026 – Present **Moodloop —** Mobile personal informatics
Designed a mobile app and researcher companion for numeric mental-state reporting, with configurable scales and multimodal contextual annotations (images, audio, tags, drawings).
Deployed a 42-day field study with 15 completing participants (594 reports; 45 interviews) and analyzed how people construct, calibrate, and contextualize scores over time.

Current **CareWork —** AI-assisted clinical dashboard
Designed an AI-assisted clinician dashboard that connects care goals, between-session tasks, patient-reported evidence, evaluation, and plan revision (HomeWork create–evaluate–iterate cycle).
Conducted formative interviews with 6 healthcare professionals and prototype sessions across 11 evaluations (14 unique HCPs) to refine usability and workflow fit.

2026 – Present **EcoCare —** Human-centered AI visualization
Proposing an AI-assisted interactive visualization of the chronic diabetes care ecology so patients, caregivers, and clinicians can inspect how life-changing events ripple across people, practices, and technologies.
Grounding the design in 22 foundational patient interviews; planning a mixed-methods study across three stakeholder groups.

2022 – 2025 **PECSS —** Clinical interfaces for PTSD therapy
Designed and implemented clinician and patient-facing tools for Prolonged Exposure therapy with Emory clinical partners, including a Clinician Homework Review dashboard over sensor-captured exercise data.
Evaluated the CHR prototype with 10 clinicians; findings published at CSCW 2024 and informed CHI 2026 clinical decision-support work.

2022 – Present **DUCSS —** Mobile sensing for diabetes care
Leading a mobile sensing system for everyday diabetes self-management and care, in collaboration with Emory University and Grady Memorial Hospital.
Designing user studies and system architecture so patients and caregivers can interact with the system individually rather than only as dyads.

2022 – Present **Trackya —** Mobile app / contextual sensing
Built a mobile app that presents step counts alongside contextual signals (location, weather, calendar, self-reports) to support sensemaking about sedentary behavior.
Completed a 28-day field deployment with 18 participants (3 interviews each). Currently analyzing the interview data and writing the paper; early findings were presented at a CHI 2025 workshop.

2021 – 2023 **Caregiving —** Qualitative social computing
First-authored a four-phase qualitative analysis of 427 posts and 888 comments from 118 informal caregivers across five online health communities.
Developed design implications for supporting caregivers' temporal sense-making around conflicts and life-changing events (CSCW 2025).

2021 – 2022 **Planneregy —** iOS app / field study
Designed, built, and shipped an iOS app for reflective weekly physical activity planning (strategies vs. day-to-day plans).
Ran a 42-day field deployment with 16 completing participants (434 plans; 48 distinct strategies); published at CHI 2024.

2021 – 2022 **Physicify —** iOS app / field study
Designed and deployed an iOS research probe with historical summaries, calendar views, and plan-detail views for reviewing prior activity plans.
Ran a 28-day field study with 17 completing participants (248 reported plans) and multi-session interviews (CHI 2022).

Research experience

2022 – Present **Graduate Research Assistant, Ubicomp Health and Wellness Lab**
Georgia Institute of Technology
Lead end-to-end research and product design for health sensing and self-management systems (DUCSS, Trackya, Moodloop, CareWork, EcoCare), spanning study protocols, Figma UI, and mobile/web deployment.
Partner with Emory clinical stakeholders on PECSS tools that use sensor-captured patient-generated data to support Prolonged Exposure therapy decision-making.
Mentor graduate students on system development and study protocols (alumni now SDE at Apple, UX at Lucidchart, and PhD at Stevens).

2019 – 2022 **Graduate Research Assistant, School of Information**
University of Michigan
Designed, implemented, and deployed iOS research apps (Planneregy, Physicify) and ran 28- and 42-day in-the-wild studies on planning, reflection, and historical records.
Built Pace to Plan, a React Native hybrid ESM platform for low-burden reporting among older adults managing chronic fatigue; translated findings into iterative interaction changes and CHI/CSCW publications.

2021 – 2022 **Research Intern, Michigan Medicine**
University of Michigan
Designed and deployed Taperology, a clinician-facing web dashboard with a custom taper-calculator algorithm for safe, individualized benzodiazepine taper schedules.
Collaborated with clinical partners to turn workflow requirements into a usable decision-support interface.

Teaching experience

Fall 2025 – Summer 2026 **Instructor, CS8001ODW: Developing Mobile Experience for Well-Being**
Georgia Institute of Technology — Designed and developed a 16-week course on health informatics and mobile app design and **taught it for three semesters**.

Spring – Summer 2024; Fall 2026 **Graduate Teaching Assistant, CS 7470: Mobile and Ubiquitous Computing**
Georgia Institute of Technology — Led assignments and in-class activities under Prof. Thomas Ploetz and Prof. Clint Zeagler (2024) and Prof. Alexander Adams (Fall 2026).

Presentations and invited talks

2026 **Doctoral Colloquium Presentation** — *UbiComp / ISWC 2026*
2026 **Paper Presentation** — *CSCW Conference, Bergen, Norway*
2026 **Trackya Project Presentation** — *CHI 2026 Workshop*
2024 **Research Showcase** — *IPaT Research Showcase*
2023 **Poster Presentation** — *CRIDC Poster Competition*

Mentorship and service

2022 – Present **Student Mentorship**
Cynthia Baseman (PhD, GaTech); Diptark Bose (MS, now SDE at Apple); Hao Yuan (MS, now PhD at Stevens); Abhinav Thukral (MS, now UX Designer at Lucidchart); Riya Soneji (MS, GaTech).

2022 – Present **Conference Reviewer**
ACM CHI (2022, 2025, 2026, 2027), ACM IMWUT (2023, 2025), IEEE ICHI (2022), CHI Late-Breaking Work (2022).

Technical skills

Research methods: Mixed methods, experimental & field studies, usability evaluation, interviews, surveys, ESM/diary studies, thematic analysis, co-design, contextual inquiry
Design & prototyping: Interaction design, Figma, wireframing, clinical and consumer health interfaces
Programming & tools: React Native, Swift, Expo, Python, JavaScript/HTML/CSS, Java, C++, Git, Unity, Tableau